Introduction to Mathematics for Deep Learning: Homeworks #1 Solutions

1. For any function $f : \mathcal{X} \to \mathbb{R}$, show that

$$\mathbb{E}\left[f(X)\right] = \sum_{x \in \mathcal{X}} f(x) \cdot p_X(x). \tag{1}$$

Hint: You can set a new random variable $Y = f(X)$.

Solution:
Let $Y = f(X)$.

$$\begin{aligned}
\mathbb{E}\left[f(X)\right] =& \mathbb{E}\left[Y\right] && (2)\\
=& \sum_y y p_Y(y) && (3)\\
=& \sum_y \sum_{x: f(x)=y} y p_X(x) && (4)\\
=& \sum_y \sum_{x: f(x)=y} f(X) p_X(x) && (5)\\
=& \sum_x f(X) p_X(x). && (6)
\end{aligned}$$

For a continuous random variable, the proof follows similarly by using integrals, so it's sufficient to demonstrate the discrete case.

2. (a) Let $X \in \{1, 2, \dots\}$ be a geometric distribution with $p = 1/2$. I.e.,

$$p_X(x) = \frac{1}{2^x} \tag{7}$$

for $x \geq 1$. What is the entropy of X?

(b) Find a discrete random variable $X \in \{1, 2, \dots\}$ that has infinite entropy.

Solution:

(a) We have

$$\begin{aligned}
H(X) =& -\sum_{x=1}^{\infty} \frac{1}{2^x} \log_2 \frac{1}{2^x} && (8)\\
=& \sum_{x=1}^{\infty} x \frac{1}{2^x} && (9)\\
=& 2. && (10)
\end{aligned}$$

(b) Let X be a random variable with pmf

$$p_X(x) = C\frac{1}{x\log^2 x} \tag{11}$$

for $x \geq 2$ and for some normalization factor $C > 0$. Note that it is a valid pmf since the sum of pmf is bounded.

$$\sum_{x=2}^{\infty} \frac{1}{x\log^2 x} \leq \frac{1}{2\log^2 2} + \int_2^{\infty} \frac{1}{x\log^2 x}\, dx < \infty. \tag{12}$$

The entropy of X is given by

$$\begin{aligned}
H(X) =& \sum_{x=2}^{\infty} C\frac{1}{x\log^2 x}\log(x\log^2 x/C) && (13)\\
=& \sum_{x=2}^{\infty} C\frac{1}{x\log x} + \sum_{x=2}^{\infty} C\frac{\log(\log^2 xC)}{x\log^2 x} && (14)\\
\geq& \sum_{x=2}^{\infty} C\frac{1}{x\log x} && (15)\\
\geq& \int_2^{\infty} C\frac{1}{x\log x}\, dx && (16)\\
=& \infty. && (17)
\end{aligned}$$

The final equality is due to $\int \frac{1}{x\log x}\, dx = \log\log x + K$ which is an increasing function.

3. Let X be a Bernoulli random variable with parameter p. Then, we define a binary entropy function $h_2(p) = H(X) = -p\log p - (1-p)\log(1-p)$. Show that $h_2(p)$ is concave.

Solution:
We have $h_2'(p) = \log\frac{1-p}{p}$ and $h_2''(p) = -\frac{1}{p} - \frac{1}{1-p} < 0$ which implies $h_2'(p)$ is decreasing. Thus, for $0 \leq x < y \leq 1$, by mean value theorem, there exists $x \leq z \leq y$ such that

$$\frac{h_2(x) - h_2(y)}{x - y} = h_2'(z). \tag{18}$$

Then, for $0 < p_1 < p_2 \leq 1$ and $0 \leq \lambda \leq 1$, we have

$$\begin{aligned}
&\lambda h_2(p_1) + (1-\lambda)h_2(p_2) - h_2(\lambda p_1 + (1-\lambda)p_2)) && (19)\\
=&\lambda(h_2(p_1) - h_2(\lambda p_1 + (1-\lambda)p_2)) + (1-\lambda)(h_2(p_2) - h_2(\lambda p_1 + (1-\lambda)p_2)) && (20)\\
=&\lambda(p_1 - (\lambda p_1 + (1-\lambda)p_2))h_2'(q_1) + (1-\lambda)(p_2 - (\lambda p_1 + (1-\lambda)p_2))h_2'(q_2) && (21)
\end{aligned}$$

for some q_1, q_2 such that $p_1 \leq q_1 \leq \lambda p_1 + (1-\lambda)p_2 \leq q_2 \leq p_2$. Thus, we have

$$\lambda h_2(p_1) + (1-\lambda)h_2(p_2) - h_2(\lambda p_1 + (1-\lambda)p_2)) \tag{22}$$

$$=\lambda(1-\lambda)(p_1-p_2)h_2'(q_1)-\lambda(1-\lambda)(p_1-p_2)h_2'(q_2) \tag{23}$$
$$=\lambda(1-\lambda)(p_1-p_2)(h_2'(q_1)-h_2'(q_2)) \tag{24}$$
$$\leq 0. \tag{25}$$

Thus, the binary entropy function $h_2(p)$ is concave.

4. The proof presented in class for $D(P\|Q) \geq 0$ is incomplete due to an edge case where $p_X(x) = 0$ for some x. Explain why the previous proof is incomplete if there exists an x such that $p_X(x) = 0$, and provide a rigorous proof that $D(P\|Q) \geq 0$ holds even in this case.

Solution:
The rigorous proof is

$$H(X) - \mathbb{E}\left[\log \frac{1}{Q(X)}\right] = \sum_x P_X(x) \log \frac{Q(x)}{P_X(x)} \tag{26}$$
$$= \sum_{x:P_X(x)>0} P_X(x) \log \frac{Q(x)}{P_X(x)} \tag{27}$$
$$\leq \log \sum_{x:P_X(x)>0} P_X(x) \frac{Q(x)}{P_X(x)} \tag{28}$$
$$= \log \sum_{x:P_X(x)>0} Q(x) \tag{29}$$
$$\leq \log 1 = 0. \tag{30}$$

To achieve equality, both the first (Jensen) and the second ($\sum_{x:P_X(x)>0} Q(x) \leq 1$) inequality should hold with equality. In other words, 1) $Q(x)/P_X(x) = c$ is constant for all x and 2) $Q(x) = 0$ if $P_X(x) = 0$. This can only be true if $Q = P_X$.

5. (a) Consider jointly distributed random variables X and Y. For any functions $f : \mathcal{X} \to \mathbb{R}$ and $g : \mathcal{Y} \to \mathbb{R}$, show that

$$\mathbb{E}\left[f(X) + g(Y)\right] = \mathbb{E}\left[f(X)\right] + \mathbb{E}\left[g(Y)\right]. \tag{31}$$

(b) Let X and Y be independent random variables. For any functions $f : \mathcal{X} \to \mathbb{R}$ and $g : \mathcal{Y} \to \mathbb{R}$, show that

$$\mathbb{E}_{X,Y}\left[f(X)g(Y)\right] = \mathbb{E}_X\left[f(X)\right]\mathbb{E}_Y\left[g(Y)\right]. \tag{32}$$

Solution:

(a) We have

$$\sum_{x,y} p_{X,Y}(x,y)(f(x)+g(y)) = \sum_{x,y} p_{X,Y}(x,y)f(x) + \sum_{x,y} p_{X,Y}(x,y)g(y) \tag{33}$$

$$= \sum_x p_X(x) f(x) + \sum_y p_Y(y) g(y) \tag{34}$$

$$= \mathbb{E}\left[f(X)\right] + \mathbb{E}\left[g(Y)\right]. \tag{35}$$

(b) We have

$$\sum_{x,y} p_{X,Y}(x,y) f(x) g(y) = \sum_x \sum_y p_X(x) p_Y(y) f(x) g(y) \tag{36}$$

$$= \sum_x \left(p_X(x) f(x) \sum_y p_Y(y) g(y) \right) \tag{37}$$

$$= \sum_x p_X(x) f(x) \mathbb{E}\left[g(Y)\right] \tag{38}$$

$$= \mathbb{E}\left[f(X)\right] \mathbb{E}\left[g(Y)\right]. \tag{39}$$

For a continuous random variable, the proof follows similarly by using integrals, so it's sufficient to demonstrate the discrete case.

6. Let X and Y be a joint random variable with pmf $p_{X,Y}(x,y)$. If there exists $\phi_1 : \mathcal{X} \to \mathbb{R}$ and $\phi_2 : \mathcal{Y} \to \mathbb{R}$ such that $p_{X,Y}(x,y) = \phi_1(x)\phi_2(y)$ for all x and y, show that X and Y are independent.

Solution:
If we marginalize $p_{X,Y}$, we have

$$p_X(x) = \sum_y p_{x,y}(x,y) \tag{40}$$

$$= \sum_y \phi_1(x)\phi_2(y) \tag{41}$$

$$= C_1 \phi_1(x) \tag{42}$$

where $C_1 = \sum_y \phi_2(y)$. Similarly, we have $p_Y(y) = C_2 \phi_2(y)$ where $C_2 = \sum_x p_X(x)$. Thus, $p_{X,Y}(x,y) = C_1 C_2 p_X(x) p_Y(y)$ where $C_1 C_2$ should be 1, which implies that X and Y are independent.

7. **Entropy of Hamming Code.** Hamming code is a simple error-correcting code that can correct up to one error in a sequence of bits. Now consider information bits $X_1, X_2, X_3, X_4 \in \{0,1\}$ chosen uniformly at random, together with check bits X_5, X_6, X_7 chosen to satisfy the following parity rules.

$$X_1 + X_2 + X_3 + X_5 = 0 \bmod 2 \tag{43}$$

$$X_1 + X_2 + X_4 + X_7 = 0 \bmod 2 \tag{44}$$

$$X_1 + X_3 + X_4 + X_6 = 0 \bmod 2 \tag{45}$$

Thus, for example, That is, $X_1 X_2 X_3 X_4 = 1011$ becomes $X_1 X_2 X_3 X_4 X_5 X_6 X_7 = 1011010$.

(a) What is the entropy $H(X_1, X_2, \ldots, X_7)$?

Now we make an error (or not) in one of the bits (or none). Let $Y = X$ XOR e, where e is equally likely to be $(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, 0, \ldots, 0, 1), (0, 0, \ldots, 0)$ and e is independent of X.

(b) Show that one can recover the message X perfectly from Y. (Please provide a justification, detailed proof not required.)

(c) What is $H(X|Y)$?

(d) What is $I(X;Y)$?

(e) What is the entropy of Y?

Solution:

(a) By the chain rule,

$$H(X_1, X_2, ..., X_7) = H(X_1, X_2, X_3, X_4) + H(X_5, X_6, X_7|X_1, X_2, X_3, X_4).$$

Since X_5, X_6, X_7 are all deterministic functions of X_1, X_2, X_3, X_4, we have

$$H(X_5, X_6, X_7|X_1, X_2, X_3, X_4) = 0.$$

And since X_1, X_2, X_3, X_4 are independent Bernoulli(1/2) random variables,

$$H(X_1, X_2, ..., X_7) = H(X_1) + H(X_2) + H(X_3) + H(X_4) = 4.$$

(b) We will expand $H(\mathbf{X} \oplus \mathbf{e}, \mathbf{X})$ in two different ways, using the chain rule. On one hand, we can write

$$\begin{aligned} H(\mathbf{X} \oplus \mathbf{e}, \mathbf{X}) &= H(\mathbf{X} \oplus \mathbf{e}) + H(\mathbf{X}|\mathbf{X} \oplus \mathbf{e}) \\ &= H(\mathbf{X} \oplus \mathbf{e}). \end{aligned}$$

In the last step, $H(\mathbf{X}|\mathbf{X} \oplus \mathbf{e}) = 0$ because $\mathbf{X}$ is a deterministic function of $\mathbf{X} \oplus \mathbf{e}$, since the (7,4) Hamming code can correctly decode $\mathbf{X}$ when there is at most one error. (You can check this by experimenting with all possible error patterns satisfying this constraint.)

On the other hand, we can also expand $H(\mathbf{X} \oplus \mathbf{e}, \mathbf{X})$ as follows:

$$\begin{aligned} H(\mathbf{X} \oplus \mathbf{e}, \mathbf{X}) &= H(\mathbf{X}) + H(\mathbf{X} \oplus \mathbf{e}|\mathbf{X}) \\ &= H(\mathbf{X}) + H(\mathbf{X} \oplus \mathbf{e} \oplus \mathbf{X}|\mathbf{X}) \\ &= H(\mathbf{X}) + H(\mathbf{e}|\mathbf{X}) \\ &= H(\mathbf{X}) + H(\mathbf{e}) \\ &= 4 + H(\mathbf{e}) \\ &= 4 + \log_2 8 \end{aligned}$$

$$= 7.$$

The second equality follows since XORing with $\mathbf{X}$ is a one-to-one function. The third equality follows from the well-known property of XOR that $y \oplus y = 0$. The fourth equality follows since the error vector $\mathbf{e}$ is independent of $\mathbf{X}$. The fifth equality follows since from part (a), we know that $H(\mathbf{X}) = 4$. The sixth equality follows since $\mathbf{e}$ is uniformly distributed over eight possible values: either there is an error in one of seven positions, or no error at all.

Equating our two different expansions for $H(\mathbf{X} \oplus \mathbf{e}, \mathbf{X})$, we have

$$H(\mathbf{X} \oplus \mathbf{e}, \mathbf{X}) = H(\mathbf{X} \oplus \mathbf{e}) = 7.$$

The entropy of $\mathbf{Y} = \mathbf{X} \oplus \mathbf{e}$ is 7 bits.

(c) As mentioned before, $\mathbf{X}$ is a deterministic function of $\mathbf{X} \oplus \mathbf{e}$, since the (7,4) Hamming code can correctly decode $\mathbf{X}$ when there is at most one error. So $H(\mathbf{X}|\mathbf{Y}) = 0$.

(d) $I(\mathbf{X}; \mathbf{Y}) = H(\mathbf{X}) - H(\mathbf{X}|\mathbf{Y}) = H(\mathbf{X}) = 4$.

8. **Conditional entropy.** Let X_1, X_2, X_3 be i.i.d. discrete random variables. Which is larger: $H(X_1|X_1 + X_2 + X_3)$ or $H(X_1 + X_2|X_1 + X_2 + X_3)$?

Solution : Conditional entropy.
They are equal.
$H(X_1|X_1 + X_2 + X_3) = H(X_1 + X_2|X_1 + X_2 + X_3)$.
proof:

$$\begin{aligned} H(X_1|X_1 + X_2 + X_3) &\overset{(a)}{=} H(X_2 + X_3|X_1 + X_2 + X_3) \\ &\overset{(b)}{=} H(X_1 + X_2|X_1 + X_2 + X_3) \end{aligned}$$

(a) follows because, given $X_1 + X_2 + X_3$, $X_2 + X_3$ is a one-one and invertible function of X_1.
(b) follows by symmetry since X_1, X_2, X_3 are i.i.d.
Many people wrote $H(X_1|X_1+X_2+X_3) = H(X_2+X_3)$, missing the conditioning on the right-hand side. This is not true in general. They ended up with wrong conclusions.